

Manual test cases suite

WebApplication

Login & Authentication Functionality

TC01: Display Login Page for Unauthenticated User

Preconditions:

- User is not logged in.

Steps:

1. Navigate to the application URL.

Expected Result:

- The login page is displayed with username and password fields.

TC0 2: Successful Login with Valid Credentials

Preconditions:

- User exists in the system with valid username and password.

Steps:

1. Open the login page.
2. Enter a valid username.
3. Enter a valid password.
4. Click the Login button.

Expected Result:

- User is authenticated successfully.
- The functionality page is displayed.

TC03: Login Attempt with Invalid Credentials

Preconditions:

- User is not logged in.

Steps:

1. Open the login page.
2. Enter an invalid username or password.
3. Click the Login button.

Expected Result:

- Login fails.
- An error message is displayed.
- User remains on the login page.

TC04: Error Message Does Not Reveal Credential Details**Preconditions:**

- User is not logged in.

Steps:

1. Attempt login with an incorrect username.
2. Attempt login with an incorrect password.

Expected Result:

- A generic error message is shown.
- The message does not specify whether the username or password is incorrect.

TC05: Access Protected Page Without Authentication**Preconditions:**

- User is not logged in.

Steps:

1. Enter the direct URL of the functionality page in the browser.

Expected Result:

- User is redirected to the login page.

Logout Functionality

TC06: Logout from Authenticated Session

Preconditions:

- User is logged in.

Steps:

1. Click the Logout option.

Expected Result:

- User is logged out successfully.
- A logout confirmation or login page is displayed.

TC07: Logout Option Available on All Authenticated Pages

Preconditions:

- User is logged in.

Steps:

1. Navigate through all authenticated pages.

Expected Result:

- Logout option is visible on every authenticated page.

TC08: Browser Back Button After Logout

Preconditions:

- User was logged in and has logged out.

Steps:

1. Click the browser Back button.

Expected Result:

- Protected content is not displayed.
- User is redirected to the login page.

Remember Me Functionality

TC09: Login with "Remember Me" Enabled

Preconditions:

- User is not logged in.

Steps:

1. Open the login page.
2. Enter valid username and password.
3. Select the "Remember Me" checkbox.
4. Click Login.

Expected Result:

- User is logged in successfully.
- Remember Me preference is saved.

Test Case 10: Auto Login with Remember Me Enabled

Preconditions:

- User previously logged in with Remember Me enabled.
- Browser was closed.

Steps:

1. Reopen the browser.
2. Navigate to the application.

Expected Result:

- User is automatically logged in without entering credentials.

TC11: Login Without Remember Me

Preconditions:

- User is not logged in.

Steps:

1. Login with valid credentials without selecting Remember Me.
2. Close and reopen the browser.
3. Navigate to the application.

Expected Result:

- Login page is displayed.
- Credentials are required again.

TC12: Logout Clears Remember Me Session

Preconditions:

- User is logged in with Remember Me enabled.

Steps:

1. Click Logout.
2. Close and reopen the browser.
3. Navigate to the application.

Expected Result:

- Login page is displayed.
- Automatic login does not occur.

Session Timeout (Inactivity) Functionality

TC13: Automatic Logout After Inactivity

Preconditions:

- User is logged in.

Steps:

1. Remain inactive for more than 20 minutes.
2. Attempt any action or page navigation.

Expected Result:

- User session expires.
- User is redirected to the login page.

TC14: Session Still Active at Exactly 20 Minutes

Preconditions:

- User is logged in.

Steps:

1. Remain inactive for exactly 20 minutes.
2. Perform an action.

Expected Result:

- User remains logged in.

TC15: Session Timer Resets After User Activity

Preconditions:

- User is logged in.

Steps:

1. Remain inactive for 15 minutes.
2. Perform any action (click, navigation).
3. Remain inactive for another 15 minutes.

Expected Result:

- User remains logged in.
- Session timeout counter resets after activity.

Input Validation Functionality

TC16: Username Length Validation

Preconditions:

- User is on the login page.

Steps:

1. Enter a username shorter than 5 characters or longer than 10 characters.
2. Enter any password.
3. Click Login.

Expected Result:

- Login is rejected.
- Validation error is displayed.

TC 17: Username Character Validation**Preconditions:**

- User is on the login page.

Steps:

1. Enter a username containing special characters or spaces.
2. Enter a valid password.
3. Click Login.

Expected Result:

- Login is rejected.
- Validation error is displayed.

TC18: Password Character Validation**Preconditions:**

- User is on the login page.

Steps:

1. Enter a password containing characters outside lowercase letters (a–z).
2. Click Login.

Expected Result:

- Login is rejected.
- Validation error is displayed.

TC19: Password Length Validation**Preconditions:**

- User is on the login page.

Steps:

1. Enter a password with fewer or more than 6 characters.
2. Click Login.

Expected Result:

- Login is rejected.
- Validation error is displayed.

Stability & Performance

TC20: Application Stability with Invalid Inputs

Preconditions:

- User is on the login page.

Steps:

1. Enter random invalid data in username and password fields.
2. Click Login multiple times.

Expected Result:

- Application does not crash.
- No data corruption occurs.
- Error handling works correctly.

Server Session / Session Handling

TC21: Session Is Server-Based

Preconditions:

- User is logged in.
- Browser DevTools is available.

Steps:

1. Open DevTools → Application/Storage.
2. Identify the session cookie (or any auth token) created after login.

3. Delete the session cookie (or clear site data for the app domain).
4. Refresh the functionality page or navigate to any protected page.

Expected Result:

- The user is no longer authenticated.
- The application redirects to the login page (or requests login again).
- Protected content is not displayed.

TC22: Session Ends When Browser Is Fully Closed (If Not Using “Remember Me”)

Preconditions:

- User logs in without selecting “Remember Me”.

Steps:

1. Close all browser windows for that browser (end the session completely).
2. Reopen the browser.
3. Navigate to the application.

Expected Result:

- The login page is displayed.
- Credentials are required again.

TC23: Session Is Not Shared Across Different Browsers

Preconditions:

- User is logged in on Browser A (e.g., Chrome).
- Browser B (e.g., Firefox) has no prior login for the app.

Steps:

1. On Browser B, navigate to the application (or directly to the functionality page URL).

Expected Result:

- Browser B shows the login page.
- The authenticated session from Browser A is not reused automatically.

TC24: Session Timeout Occurs Even If Tab Stays Open (Idle > 20 Minutes)

Preconditions:

- User is logged in on the functionality page.

Steps:

1. Leave the tab open and do not interact for **more than 20 minutes**.
2. Without refreshing, click any protected action or navigate to another protected page.

Expected Result:

- The session is expired and user is forced to log in again.
- Protected content/actions are blocked.

Username Uniqueness

TC25: Reject Duplicate Username Creation (Admin UI / API)

Preconditions:

- A user already exists with username User01.
- Admin UI or API exists to create users (assumption).

Steps:

1. Attempt to create a new user with username User01 and any valid password.
2. Submit the creation request.

Expected Result:

- The system rejects the creation.
- A clear error indicates the username already exists.
- No duplicate record is created.

TC26: Database Constraint Prevents Duplicate Username

Preconditions:

- Test environment provides write access to the users table (or equivalent).
- A user already exists with username User01.

Steps:

1. Attempt to insert another user row with username User01 via approved test method (migration script, admin tool, or controlled DB insert).
2. Observe the result.

Expected Result:

- The operation fails (constraint/unique index violation) or the system blocks it.
- Only one user record exists for User01.

Performance Requirement (95% of requests < 1s)

TC27: Login Request Response Time Under 1 Second

Preconditions:

- User credentials are valid.
- Browser DevTools is available.

Steps:

1. Open DevTools → Network tab.
2. Disable cache (optional) and preserve log.
3. Perform a valid login.
4. Identify the login/auth request in the network list.
5. Capture its “Duration/Time”.

Expected Result:

- The login request completes in **< 1 second**.

TC28: Functionality Page Load Requests Under 1 Second

Preconditions:

- User is logged in.
- DevTools Network tab is open.

Steps:

1. Refresh the functionality page.
2. Identify the main document request and key API calls.
3. Record durations for each.

Expected Result:

- Key page/API requests complete in **< 1 second**

TC29: 95% Rule (Manual Sampling Approach)

Preconditions:

- Stable test environment.
- DevTools Network is available.

Steps:

1. Perform **20 repeated** login attempts (or 20 page refreshes of functionality) in a controlled manner.
2. Record the request duration each time (export HAR if possible).
3. Count how many are below 1 second.

Expected Result:

- At least **19 out of 20** (≈95%) requests are **< 1 second**.